

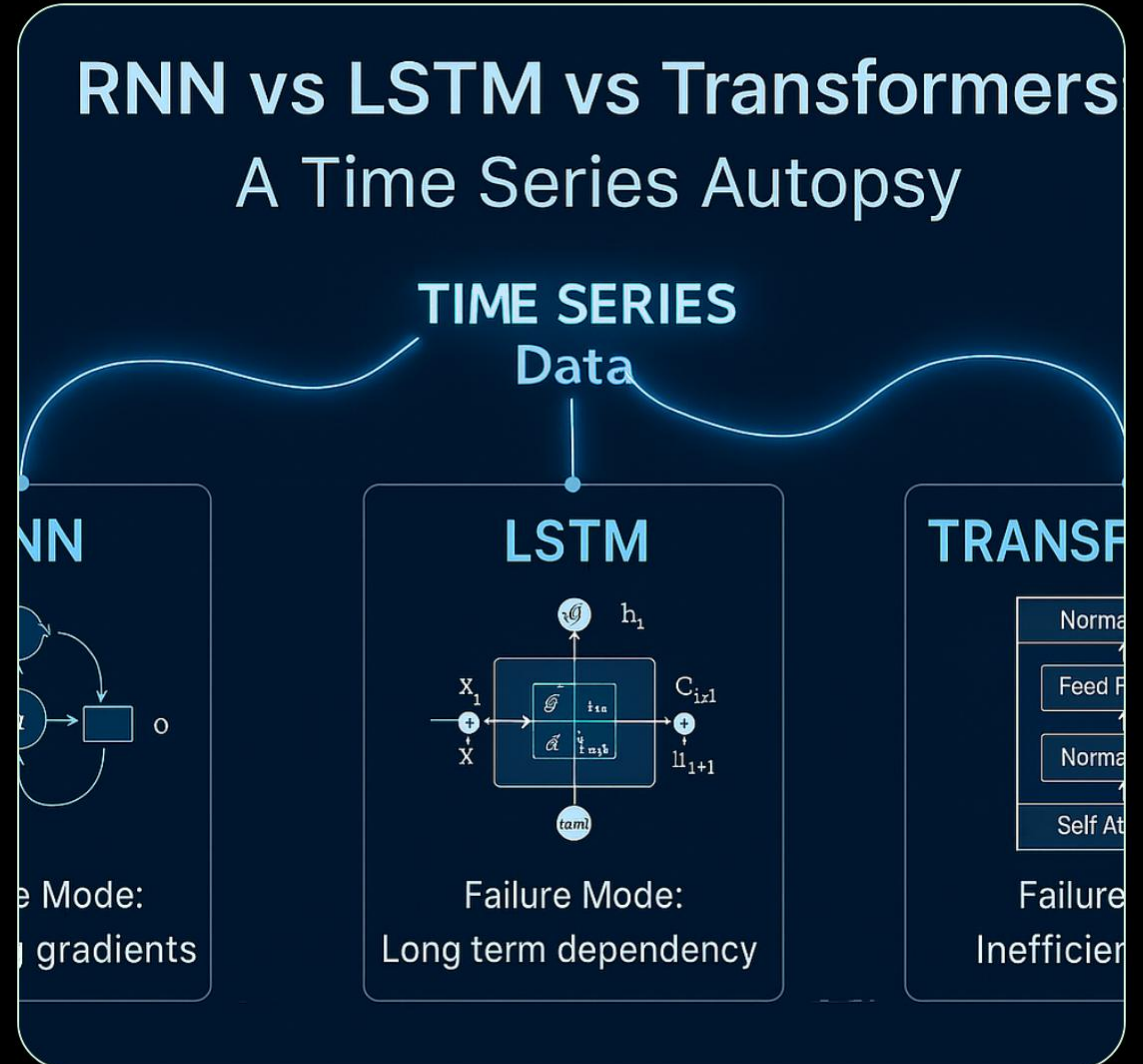
Mastering Transformers and Sentiment Analysis

Day 6: Intuition, Embeddings, and Professional Evaluation

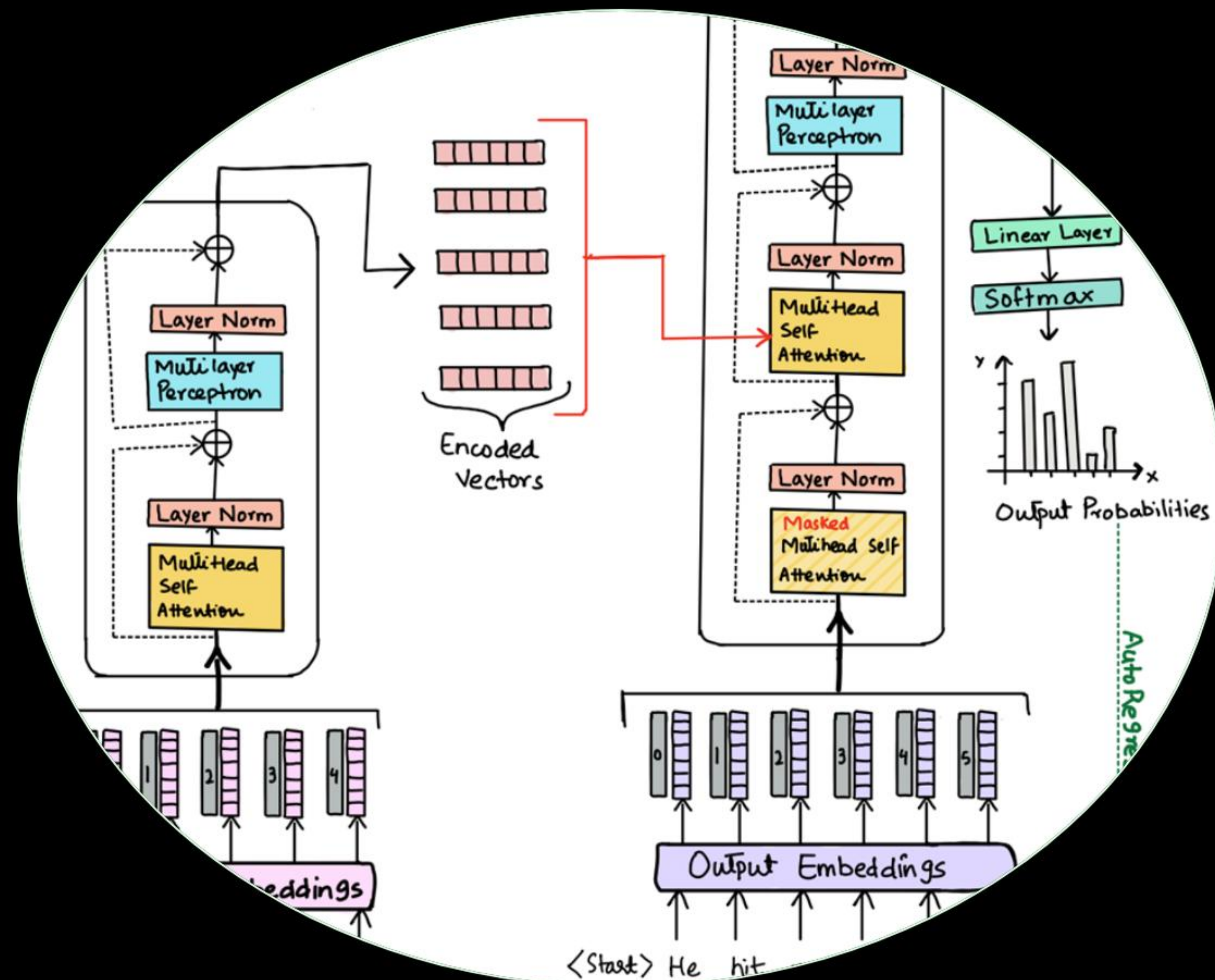
THE PARADIGM SHIFT

Why Transformers?

- **Beyond RNNs:** Moving from sequential processing to massive parallelization.
- **Long-term Dependencies:** Solving the vanishing gradient problem in long sentences.
- **The Attention Revolution:** "Paying attention" to relevant words regardless of distance.



THE SECRET SAUCE



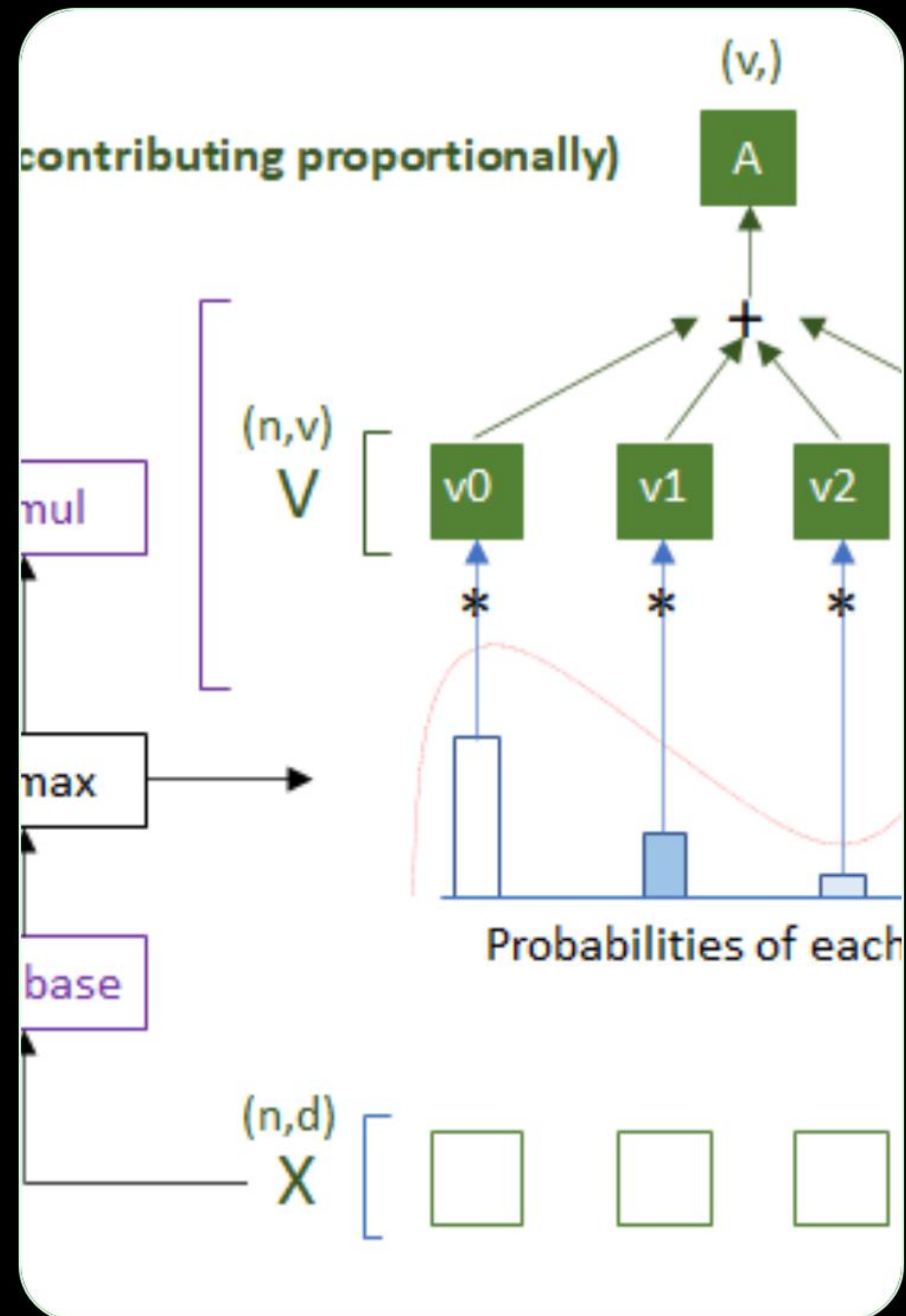
Self-Attention Intuition

Every word in a sentence looks at every other word to find its context.

*"The animal didn't cross the street because **it** was too tired."*

Attention helps the model know that **it** refers to **animal**, not **street**.

THE MATHEMATICS OF MEANING



Query (Q)

What I am looking for (the current word's request).



Key (K)

What I have to offer (how well other words match the query).



Value (V)

The actual content I provide once a match is found.

The Score

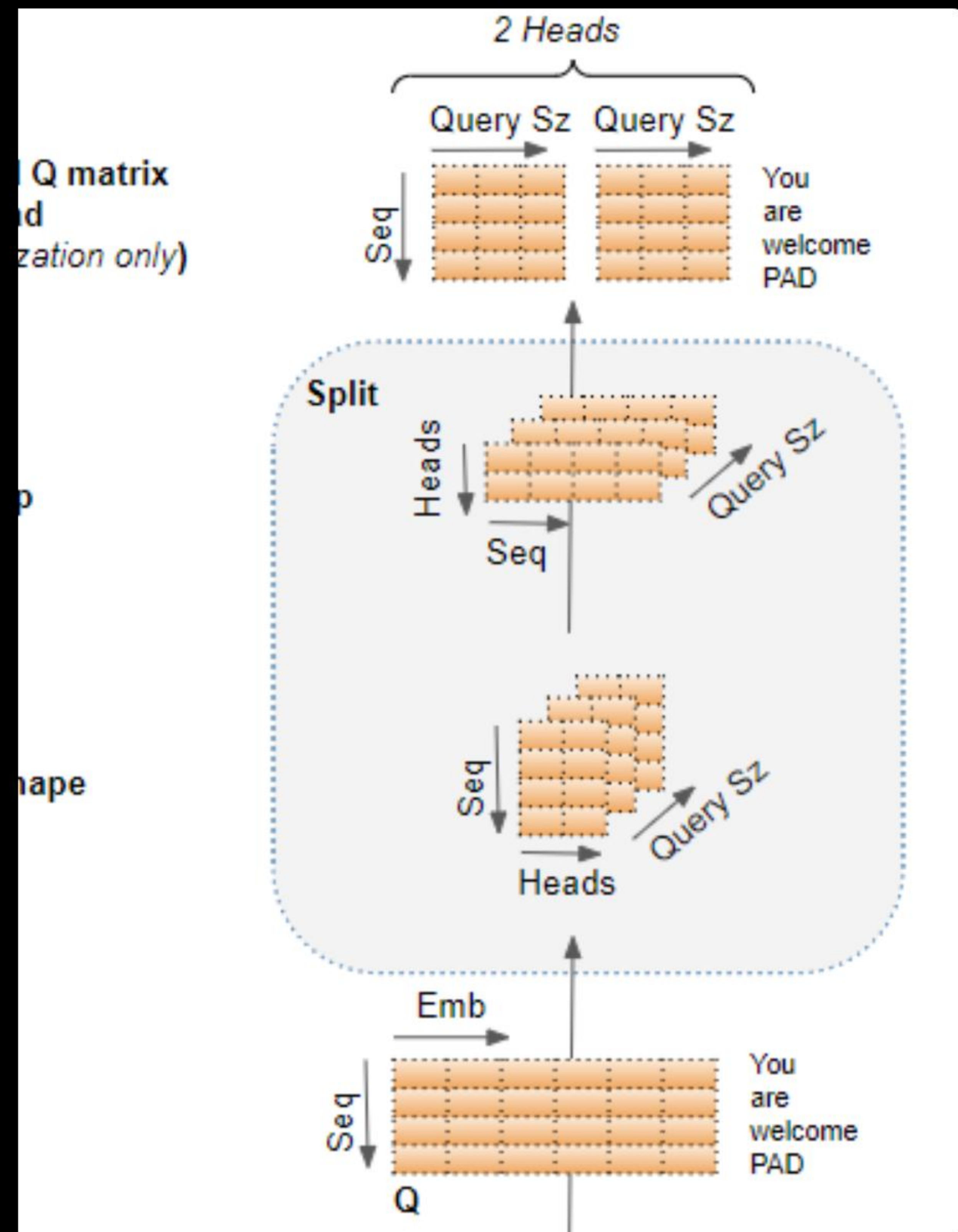
$\text{Softmax}(QK^T/\sqrt{d})V$ creates the context-aware representation.

Multi-Head Attention

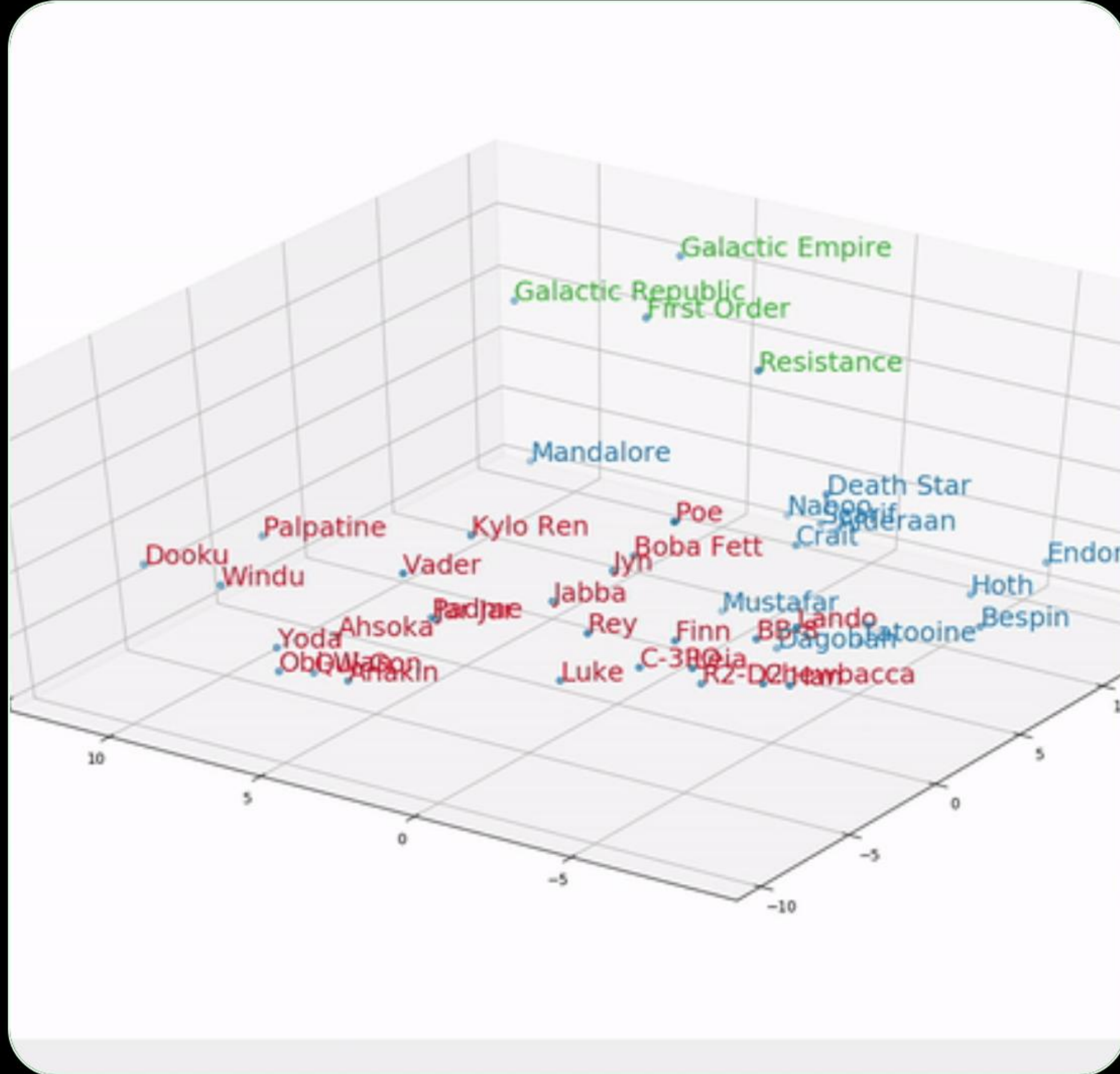
Why use one brain when you can use eight?

- **Head 1:** Focuses on grammar/syntax.
- **Head 2:** Focuses on pronouns.
- **Head 3:** Focuses on emotional sentiment.

Combining these "heads" gives the model a multi-dimensional understanding of language.



DATA REPRESENTATION



768

Standard BERT Embedding Dimensions

Text is converted into high-dimensional vectors. Words with similar meanings cluster together in this hidden space, allowing math to solve language problems.

IMPLEMENTATION STRATEGIES

Option A: Pre-trained Pipe

Use `pipeline("sentiment-analysis")` for immediate results on standard data.

```
from transformers import pipeline
```

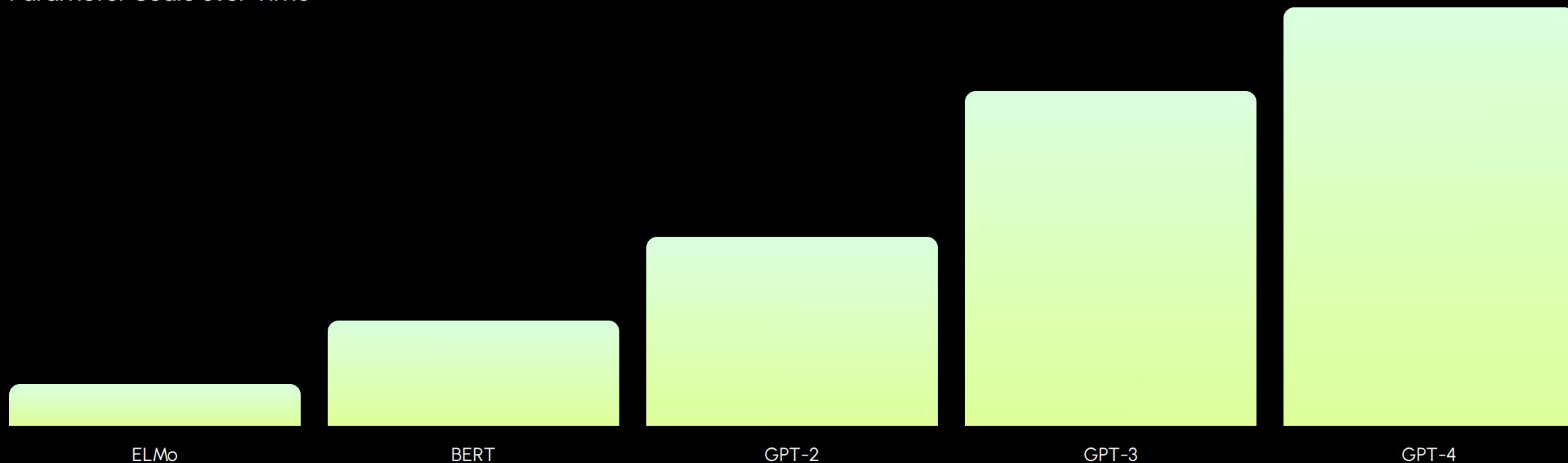
Option B: Embedding Features

Extract BERT's hidden states and feed them into your own classifier (e.g., Random Forest).

```
model = AutoModel.from_pretrained()
```

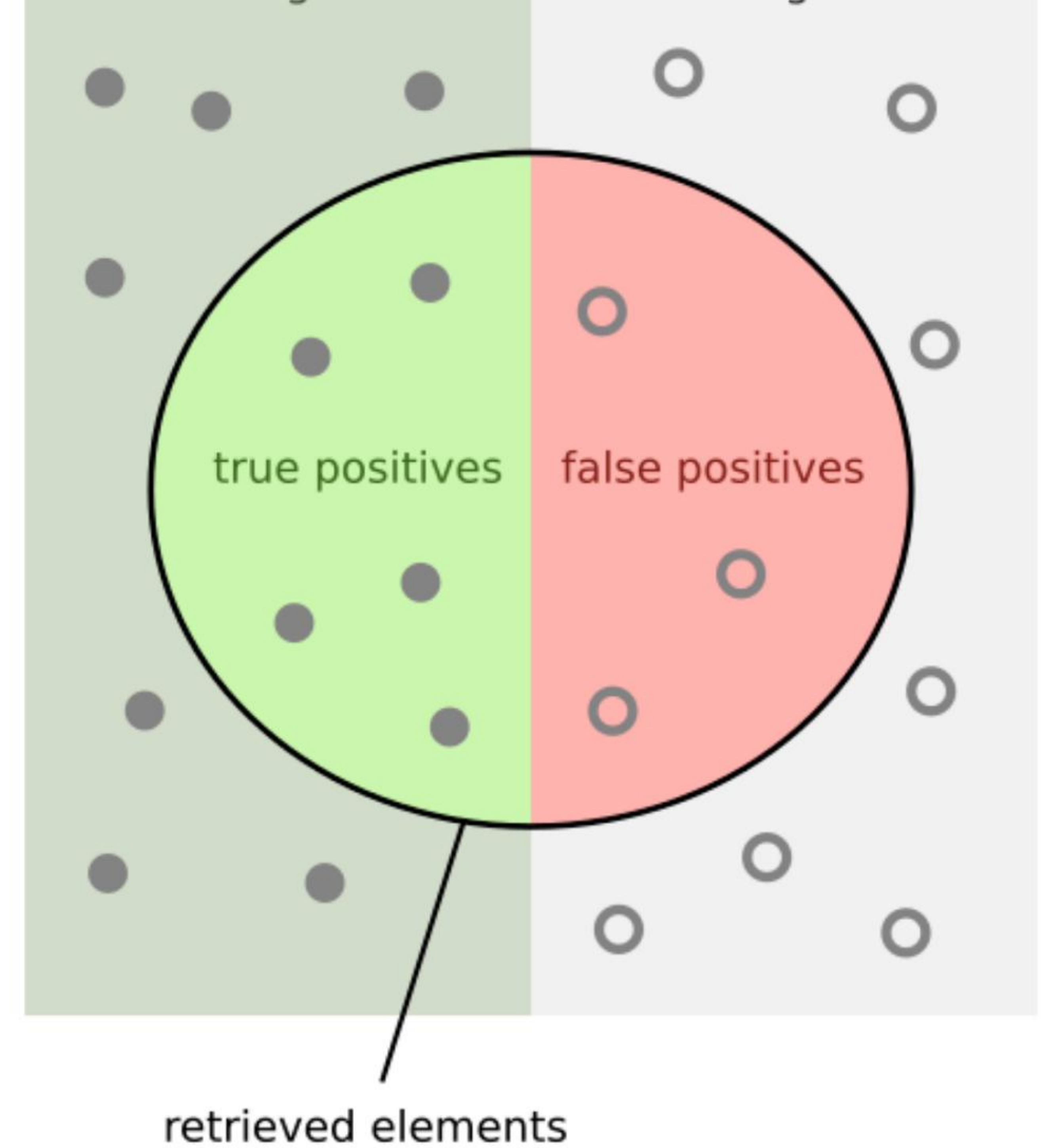

NLP CAPABILITY GROWTH

Parameter Scale over Time



The jump from sequential (RNN) to Attention-based models sparked exponential growth.

Recall	Did I catch all the positive cases?
F1-Score	Balance between Precision and Recall.



How many retrieved items are relevant?

How many relevant items are retrieved?

Conclusion & Q&A

Transformers are the building blocks of modern AI.

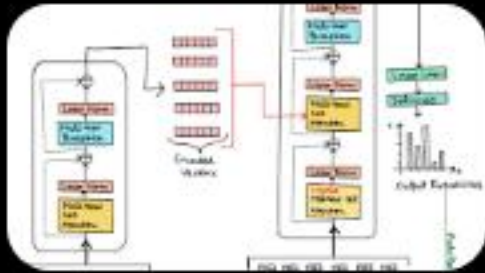
Up Next: Hands-on Fine-tuning

Image Sources



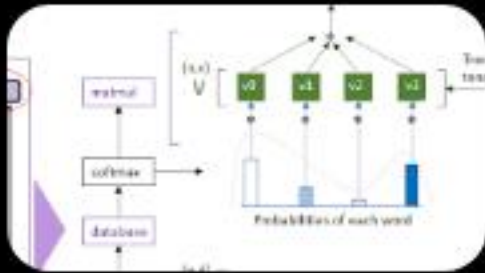
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Source: ai.plainenglish.io



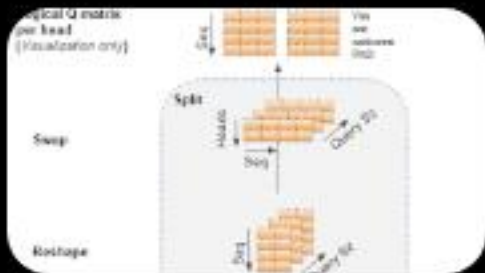
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Source: medium.com



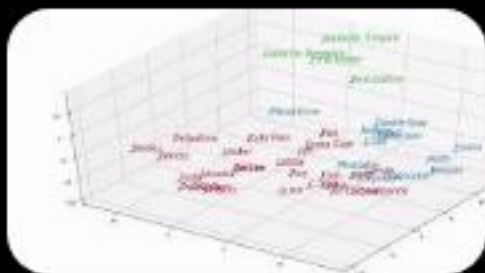
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Source: stats.stackexchange.com



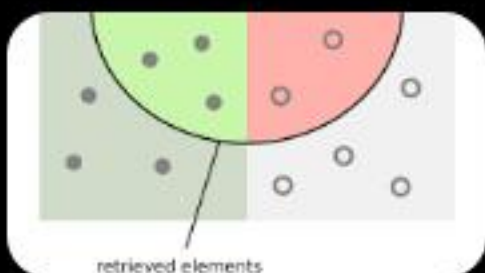
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